



Openfiler

Part—3

The first article of this series (*LFY, August 2011*) covered basic installation and configuration; the second (*LFY, September 2011*) described some advanced Openfiler features, including bonded interfaces and RAID 1 data storage using three hard disks,, with some important set-up instructions and troubleshooting tips. This article covers two important aspects of Openfiler installation—RAID 1 with two hard disks,, and accessing openfiler FTP server from the Internet using firewall port forward feature.

The Openfiler RAID 1 discussed in the September 2011 issue of *LFY* uses two hard disks for data mirroring and an additional disk for OS.. Here, to recover from a crashed OS disk, you need to replace the disk, reinstall the OS and restore back-ups. Since the RAID data volumes remained intact, the server will be back after reinstallation. The disadvantages here are the need for an additional OS disk (and its power consumption) and having to reinstall the OS to recover. It is also possible to have both OS partitions and RAID 1 volumes on just two disks, by going through the following two steps:

1. Create the required OS partitions, configure them in RAID 1, and modify GRUB to include the second hard disk as a bootable disk.
2. Create RAID 1 volumes, and assign user rights from the Web-based Openfiler GUI.

Configuration used during the test setup was an Intel Dual-Core CPU-based PC with 2 GB RAM, 2 x 500 GB SATA hard disks, a DVD drive for installation and Openfiler version 2.3.

Table 1: RAID 1 partitioning

Partition name	Type	Create in	Size in MB	Create and mount software RAID 1 array	RAID 1 array members	Purpose
Boot	Software RAID	<i>hda</i> and <i>hdb</i>	256	<i>md0</i> , mount as <i>/boot</i>	<i>hda1</i> and <i>hdb1</i>	Kernel
Root	Software RAID	<i>hda</i> and <i>hdb</i>	2048	<i>md1</i> mount as <i>/</i>	<i>hda2</i> and <i>hdb2</i>	Applications and libraries
Log	Software RAID	<i>hda</i> and <i>hdb</i>	768	<i>md2</i> mount as <i>/var/log</i>	<i>hda3</i> and <i>hdb3</i>	Logs
Swap	Swap	<i>hda</i> and <i>hdb</i>	512	N/A	N/A	Memory swap space

Start by booting from the Openfiler CD, and continue with the GUI-based installation. The most important step required for RAID 1 installation is creating the RAID partitions manually, using Disk Druid. Check Table 1 for the brief layout.

The remaining installation options are straightforward—Ethernet interface parameters (don't forget to set the interface to activate at boot), and setting the time-zone and assigning a root password. Once done, open the Web GUI from any computer in the same subnet (<https://ipaddress:446>). Verify the RAID configuration, navigate to Volumes—Software RAID and verify all the three RAID volumes created earlier. If any change is required, this is the time to fix the partitions.

Next, log in as root and update Openfiler with the following command:

```
conary updateall
conary update conary
```

Confirm that all updates have been applied, by running the same command till you receive the message: 'No new troves were found'.